EE-120 ELECTRONIC DEVICES AND CIRCUITS

COMPLEX ENGINEERING PROBLEM

Regulated Power Supply

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Abstract

In the modern era, nearly all electronic systems rely on stable and efficient power supplies. A regulated power supply is a crucial circuit that maintains a constant output voltage regardless of input fluctuations or variations in load current, ensuring the smooth and reliable operation of electronic devices. Such stability enhances device performance, prevents component damage from voltage spikes, and guarantees consistent results in measurement and control applications. This project focuses on the design and validation of a regulated DC power supply that provides a stable output of 1.7 V across a 5 kΩ load. The design incorporates a transformer-based step-down stage, bridge rectification, ripple filtering, and Zener diode voltage regulation is achieved through a 2.8V Zener diode in series with 470Ω current limiting resistor. To reach the specific target voltage, a silicon diode is placed in series with the output to utilize its inherent forward biased voltage drop of approximately 0.7V. The performance was successfully verified through both LTspice simulation and hardware experimentation.

Contents

Objectives

The main objective of this experiment is to make a 1.7V Regulated DC power Supply at the load of 5kΩ Resistive Load.

Introduction

In Pakistan, the voltage we get from our local grid stations is AC with a frequency of 50 Hz. However, most of the appliances and electric devices we use cannot directly use this AC current, rather they require a DC current which has a constant value to work properly. As the source current is AC and we require DC current, there is a need for some means which can convert that AC current into DC current. This is done by using a DC power supply. A DC power supply takes AC voltage as inputs, steps it down, filters it from any noise and finally regulates it to the required DC voltage. DC power supplies have following components:

- Step down transformer
- Rectification Circuit
- Capacitor
- Regulator
- Load Resistance

The ratings of these components depend on the output voltage required.

Theory

We provide an AC signal (220 V rms, 311.13 V peak) to a transformer. We determine the turns ratio required to step down the voltage to a required nominal value. We'll use a bridge rectifier next for full wave rectification. Considering that a rectifier and filter circuit usually drops around 1.4 to 1.6 V, so we keep in mind to keep the secondary voltage value at least 2V above of our desired output. We will use a DC filter which consists of a capacitor and a large resistor. We'll calculate the value of the capacitor knowing our peak voltage (rectified), the frequency of our AC signal and our required ripple voltage which we will assume to be minimum. Next comes our voltage regulator. We'll use a Zener diode as a voltage regulator. There is a shunt resistor in series with the Zener diode which controls the current through the diode.

DC Power Supply

A D.C power supply takes A.C source as an input and converts it into a constant D.C current to be used by the electronic equipment. As most of the appliances and electronic equipment require a constant D.C voltage for their smooth operation, D.C power supply is an essential device without which our electronic devices will fail to work properly. A DC power supply has 4 main components. A step-down transformer, which steps down the input voltage to a low value. A rectifier, which rectifies the AC voltage into DC voltage. A capacitor, which filters the rectified voltage. And finally, a regulator which regulates the voltage to a specified value.

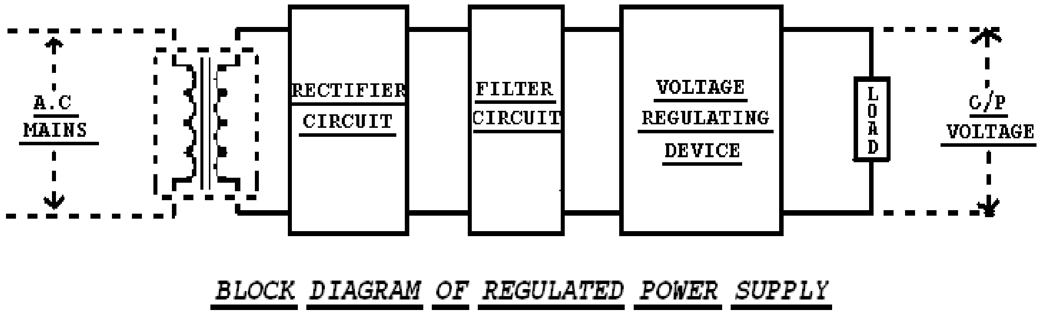

Figure 1: Block Diagram of Regulated Power Supply

Transformer

A transformer works on the principle of mutual induction, which states that when a change in magnetic flux occurs in the circuit, an E.M.F is induced in the circuit. A transformer can be either a step-up transformer or a step-down transformer, the use of which depends on the application. In a DC power supply, a step-down transformer is used, as we need to step down the 230V to some low value. In our case, we used a center taped transformer which gives 12V from any line with the neutral line and 24V from both lines.

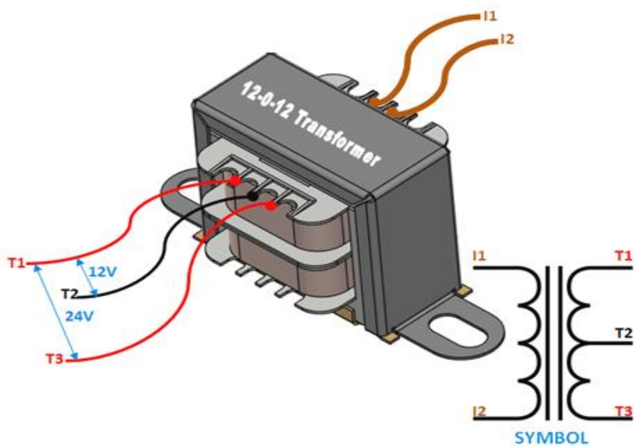

Figure 2: Step down Transformer

Rectification Circuit

A rectifier converts AC voltage into DC voltage. Rectification can be either half wave or full wave rectification. In half wave rectification, only a single diode is used in reverse bias. During the positive half cycle, it conducts current while during the negative half cycle it doesn't allow the current to flow. Thus, only positive voltage is obtained from the output. But the disadvantage of a half wave rectifier is that a lot of voltage signal is lost, and we obtain only about 30% of its value.

Full wave rectification is more efficient than half wave rectification. It uses 4 diodes in a Wheatstone bridge configuration in such a way that during positive as well as negative half cycle, 2 of the diodes are in forward bias while the other 2 are in reverse bias. Thus, only positive voltage is obtained in both cycles. In our case we used a 4 Silicon 1N4007 diode bridge rectifier.

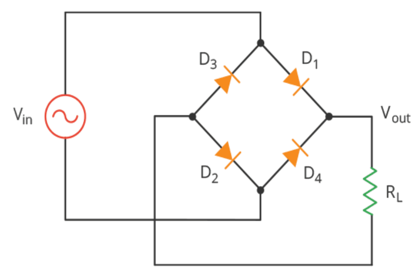

Figure 3: Bridge Rectification of 4 Diodes

Filter Circuit

The purpose of the filter circuit in a D.C power supply is to eliminate the noise in the rectified signals coming from the rectifier. The filter circuit consists of a capacitor of suitable capacitance and voltage rating by the voltages and currents in the power supply. A limiting resistor can also be used parallel to the capacitor as part of the filter circuit. The purpose is this resistance is to decrease the loading on the capacitor and to reduce the value of the required capacitance. In power supplies, capacitors are used to smooth pulsating D.C output. In our case, we used a 57µF capacitor with a voltage rating of 25V and a resistor of 10kΩ.

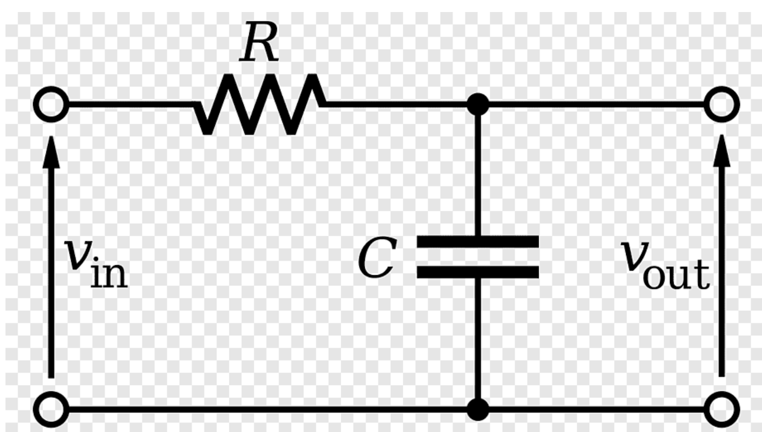

Figure 4: Filter Circuit

Regulator

A voltage regulator is used to control and stabilize the final output voltage. Because there can be sudden fluctuations in the AC mains input or changes in the load demand, the filtered DC voltage can experience jumps or drops that might harm sensitive electronic devices. To prevent this, a regulator limits the voltage to a safe, specified value. For our design, we utilized a Zener diode voltage regulator. Operating in its reverse-breakdown region, the Zener diode clamps the output to a constant, precise voltage (such as 3.3V). It is paired with a carefully calculated series resistor that limits the current, ensuring the diode operates safely while absorbing any remaining AC ripple and delivering a clean, regulated DC supply to our load.

Figure 5: Zener Diode as Voltage Regulator

Components

The Components used in making a Power Supply are

Step Down Transformer

1N4007 Diode

Capacitor

Zener Diode

Resistor

Breadboard

Connecting wire

Oscilloscope

DMM

Theoretical Design and Calculations

Transformer Design

For the transformer design we started with a standard $220V_{rms}$ AC input which translate to a peak voltage of about 311.13V since we needed a 12V output on the secondary side for our regulation stages we calculated the turn ratio $\frac{V_p}{V_s}$ to be roughly 18.3. To get this working in LTSpice we had to translate that turns ratio into inductance values since the software use the relationship $\frac{L_1}{L_2} = \left(\frac{N_1}{N_2}\right)^2$. By setting our secondary inductor L_2 to 1mH as a base we calculated that the primary inductor L_1 needed to be 336.11mH but we adjusted this to 367.3mH in our final simulation to better match our specific hardware behavior and bridge rectifier losses. This setup successfully stepped down the high voltage mains to a level where our Zener diode could safely handle the regulation.

Primary Voltage: 230V RMS and 311.13 peak Voltage measured from main power supply

Secondary Voltage: 12V RMS and 17 peak voltage measured at secondary coil of transformer

Turns Ratio: $N_p/N_s = 220/12$ = 18.33 : 1

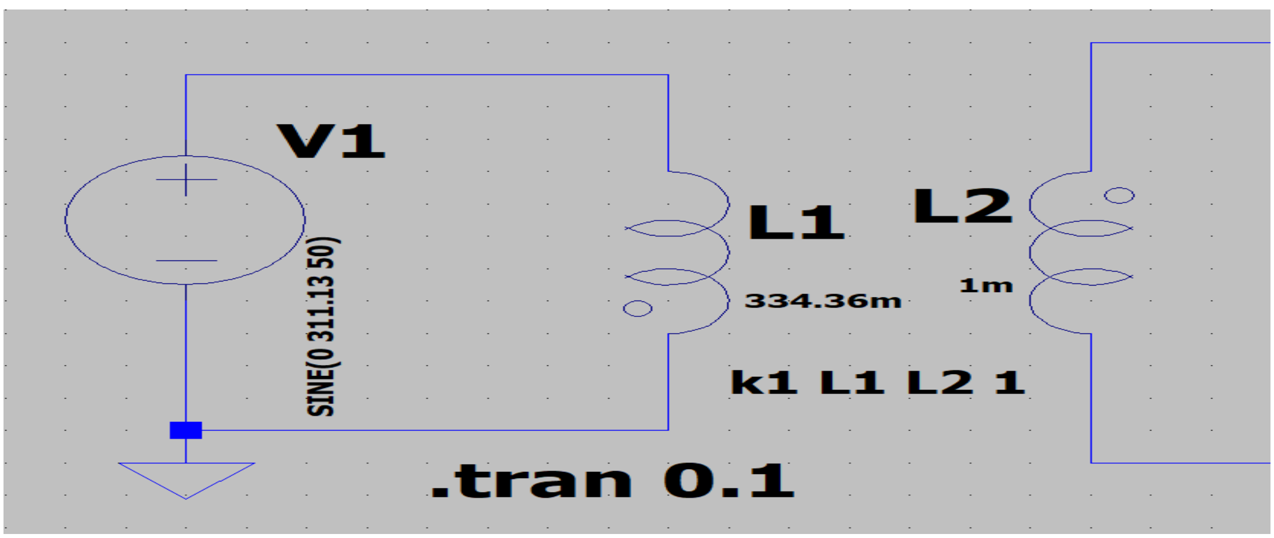

Figure 6: Transformer Setup in LTSpice

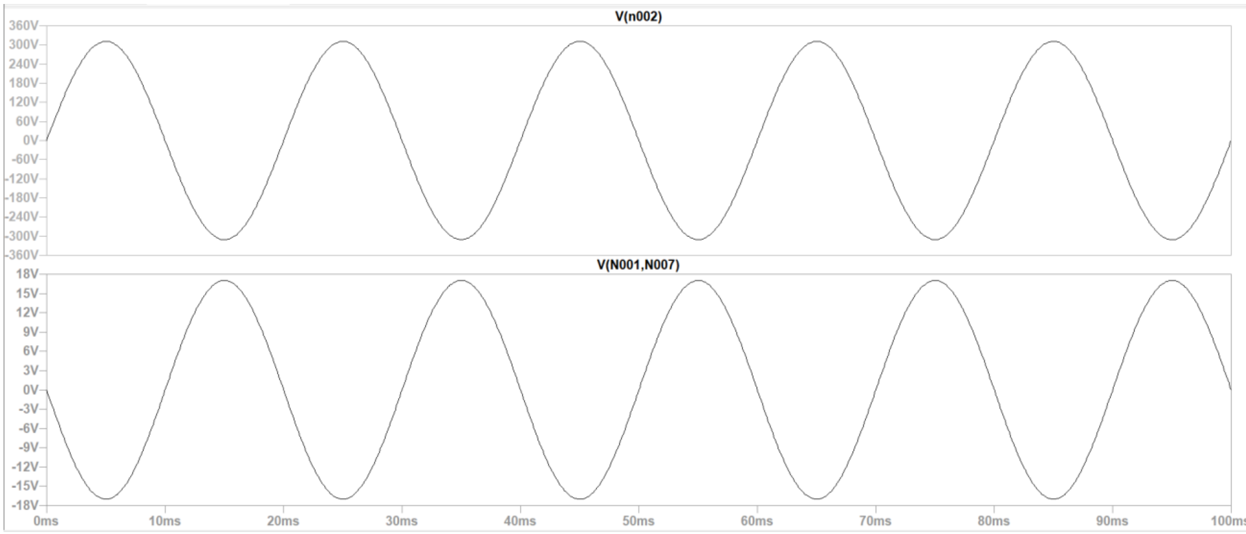

Figure 7: Output Wave across primary and secondary coil of Transformer

Bridge Rectifier Stage

For the rectification stage of the project, we implemented a full wave bridge rectifier using 1N4007 silicon diodes. This configuration was chosen to ensure that both the positive and negative halves of the $12V_{rms}$ AC secondary signal are utilized, effectively converting the alternating current into the pulsating direct current. When testing this stage with a 2.2kΩ load, we accounted for a total voltage drop of approximately 1.4V across the diode pairs.

Forward Voltage Drop: 2 × 0.7 = 1.4V

Rectified Output: $V_{rect} = 17V - 1.4V = 15.6V$

Ripple Frequency: $f_r = 2 \times 50\ Hz = 100\ Hz$

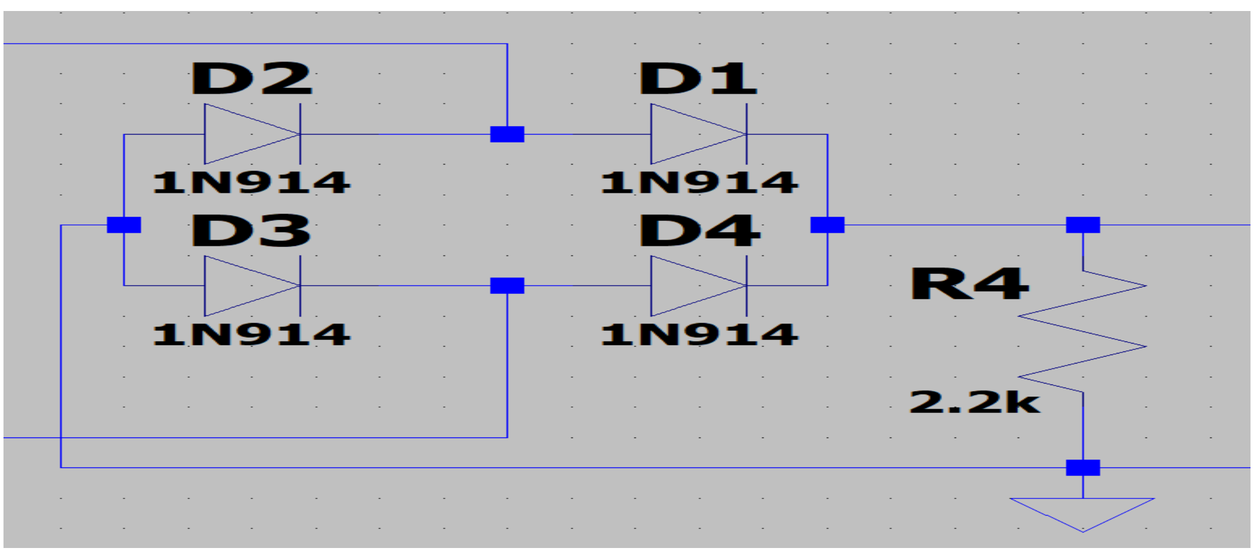

Figure 8: Bridge Rectifier in LTSpice

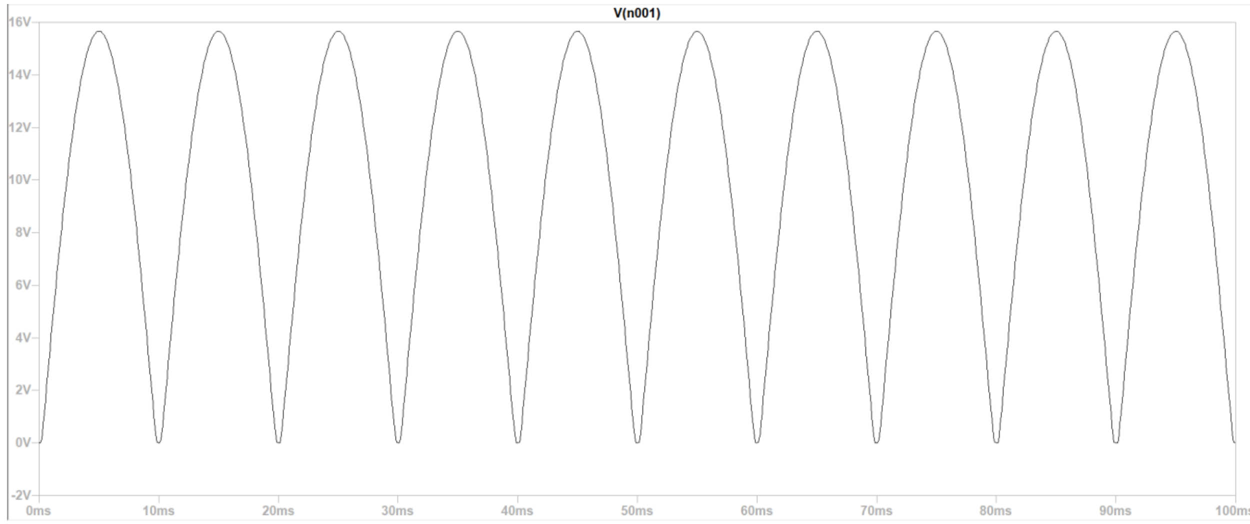

Figure 9: Output Wave across Bridge Rectifier

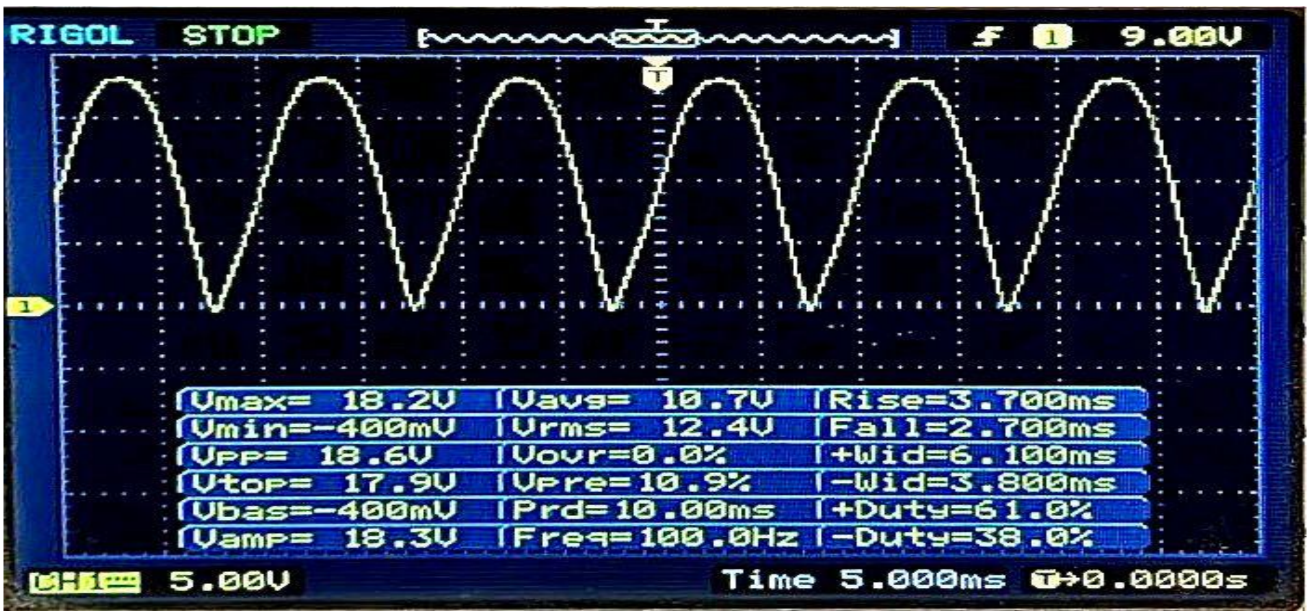

Figure 10: Output Wave across Bridge Rectifier in Oscilloscope

Filter Circuit

The filter stage is responsible for converting the pulsating DC output of the bridge rectifier into a steady, smooth DC voltage. In this project, we utilized a parallel combination of two capacitors of 47µF and 10µF. placing these capacitors in parallel results in a total equivalent capacitance of 57µF, which provides a higher storage capacity to maintain the output voltage during the interval between the rectified AC peaks.

The 47µF capacitor act as primary reservoir, smoothing out the large low frequency ripple by discharging into the load whenever the rectified input voltage drops. The additional 10µF capacitor helps to further stabilize the signal and handle the smaller, higher frequency noise. This dual capacitor arrangement ensures a low ripple factor, which is critical for the stability of the subsequent Zener regulation stage. The filter circuit provides the steady DC baseline necessary to achieve the target 1.7V output under the specified load conditions.

$$measured\ V_{pp} = 600mV$$

$$measured\ V_{avg} = 17.7V$$

$$r\% = \left(\frac{V_{pp}}{V_{avg}}\right) \times 100$$

$$r\% = \left(\frac{0.6V}{17.7V}\right) \times 100$$

$$r\% = 3.39\%$$

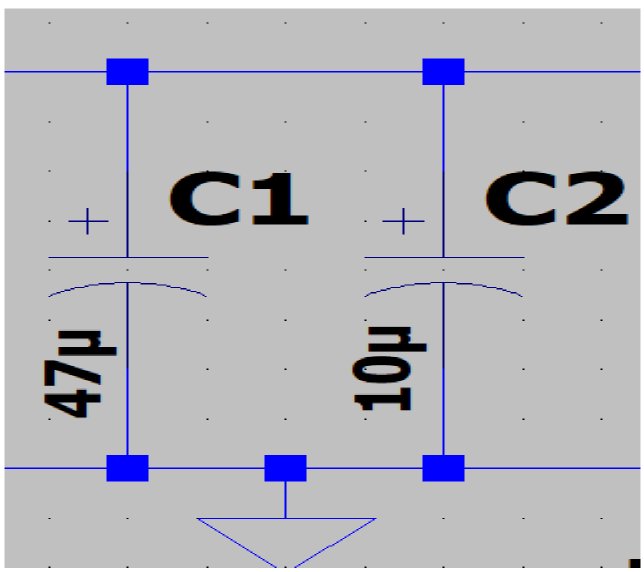

Figure 11:Filter Circuit in LTSpice

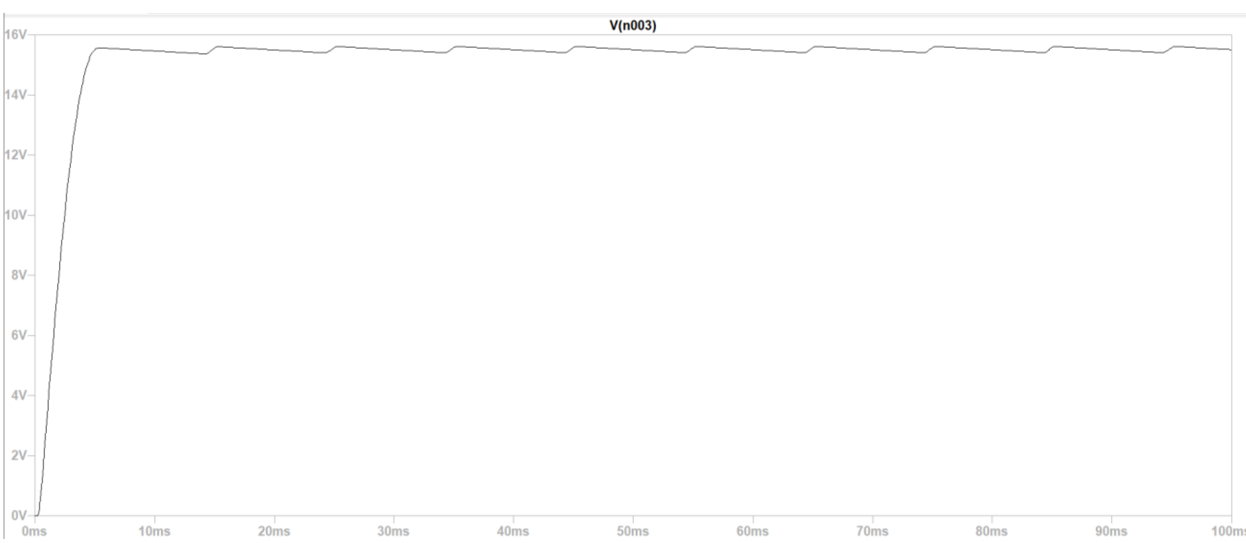

Figure 12: Output Wave across Filtering Capacitor

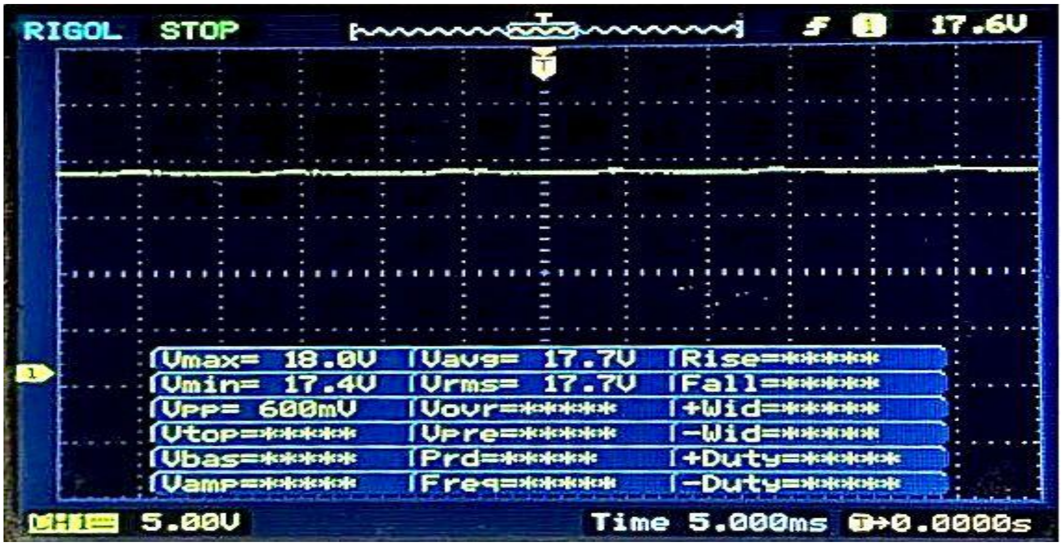

Figure 13: Output across Filtering Capacitor in Oscilloscope

Voltage Regulator

For the voltage regulation stage of the project, we utilized a Zener diode-based shunt regulator to maintain a stable output voltage. The circuit employs a 2.8V Zener diode biased through a 10kΩ series resistor which clamps the filtered DC signal to a consistent reference level. As our required design requirement was a 1.7V we incorporated a 470Ω resistor and a silicon diode in series with the load. This configuration ensure that the output voltage remain regulated and protected against minor fluctuations in the input supply, providing a reliable 1.7V DC Signal to the 5kΩ load.

$$I_L = \frac{V_{out}}{R_L} = \frac{1.7V}{5000\Omega} = 0.34\,mA$$

$$V_z = V_{R3} + V_{D6} + V_{out}$$

$$V_z = (I_L \times 470) + 0.7 + 1.7$$

$$V_z = (0.34 \times 10^{-3} \times 470) + 2.4$$

$$V_z = 2.56V$$

$$I_L = \frac{V_{in} - V_Z}{R_s}$$

$$I_L = \frac{17.7 - 2.56}{10000} = 1.514\,mA$$

$$I_z = I_S + I_L$$

$$I_z = 1.514 - 0.34 = 1.174\,mA$$

$$R_s = \frac{V_{in} - V_Z}{I_Z + I_L}$$

$$R_s = \frac{17.7 - 2.56}{(1.174 \times 10^{-3}) + (0.34 \times 10^{-3})}$$

$$R_s = 10\,K\Omega$$

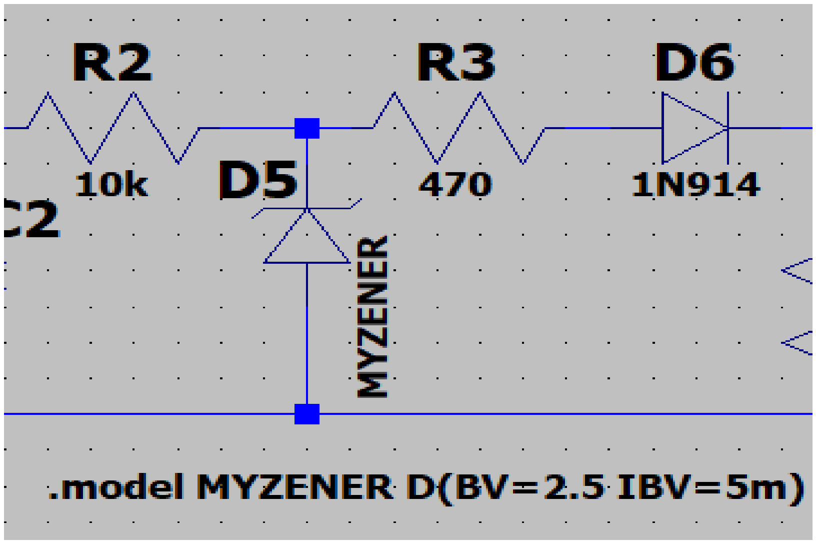

Figure 14: Zener as Voltage regulator in LTSpice

Figure 15: Output Wave across Zener Regulator in LTSpice

Output Load

For the final stage of the project, 5kΩ resistors serves as the output load, simulating the power requirements of a low voltage electronic component. With the circuit successfully regulated, the load receives a constant 1.7V, resulting in a stable load current of approximately 0.35mA. This steady performance confirms that the design effectively isolates the load from input noise and ripple, providing the precise DC environment necessary for sensitive engineering applications.

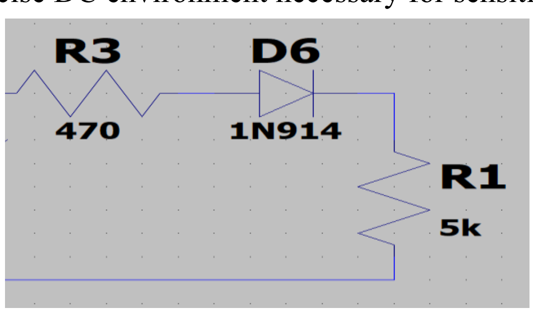

Figure 16: Load Setup in LTSpice

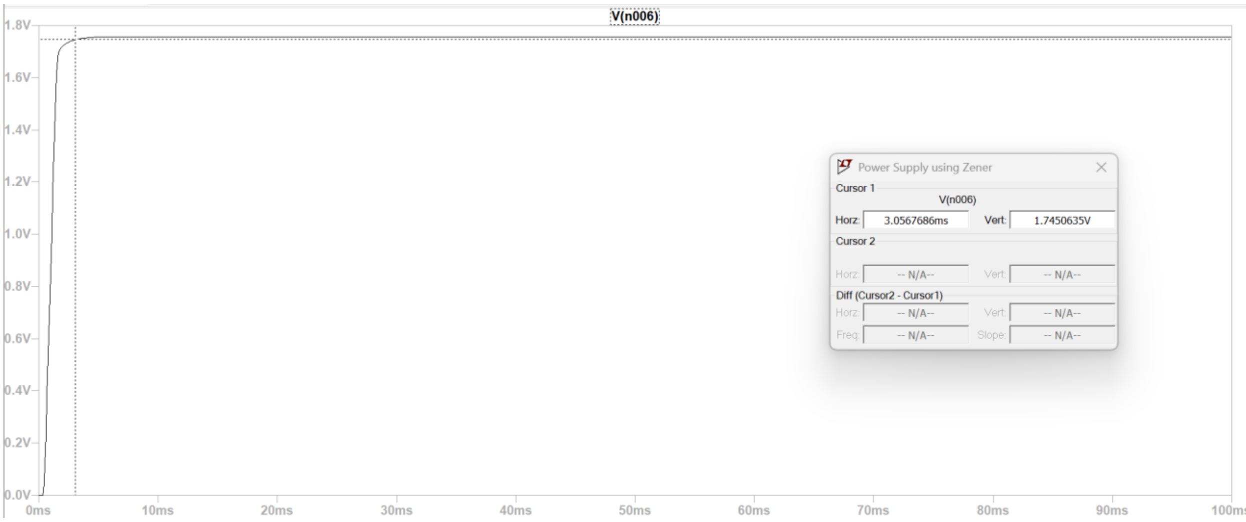

Figure 17: Output Wave across Load in LTSpice

Figure 18: Output Wave across load in Oscilloscope

Simulated circuit

We used LT Spice to make a simulation of our circuit. We used the right resistances and the right models of all the equipment available in the libraries of LT Spice and We saw from our results that a 1.7V @ 0.35mA DC power supply was made using simulation.

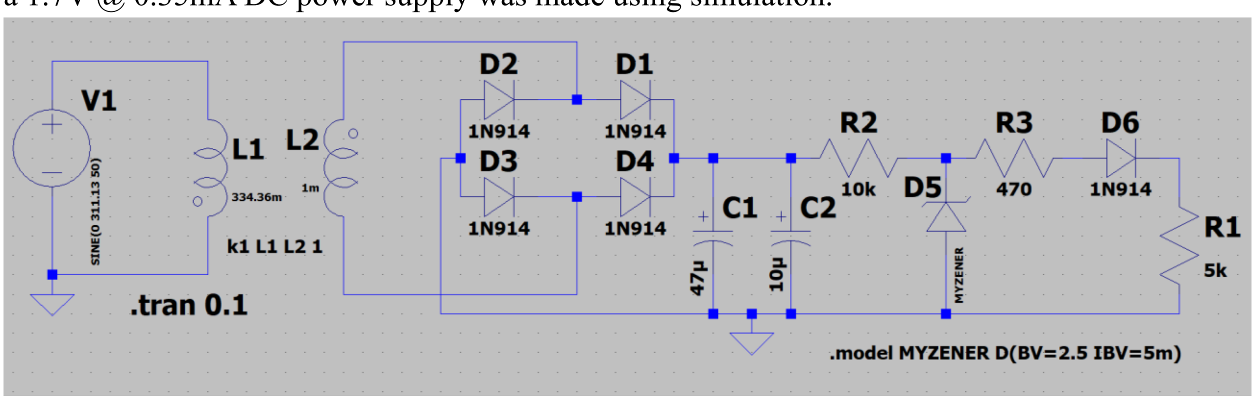

Figure 19: Complete Circuit in LTSpice

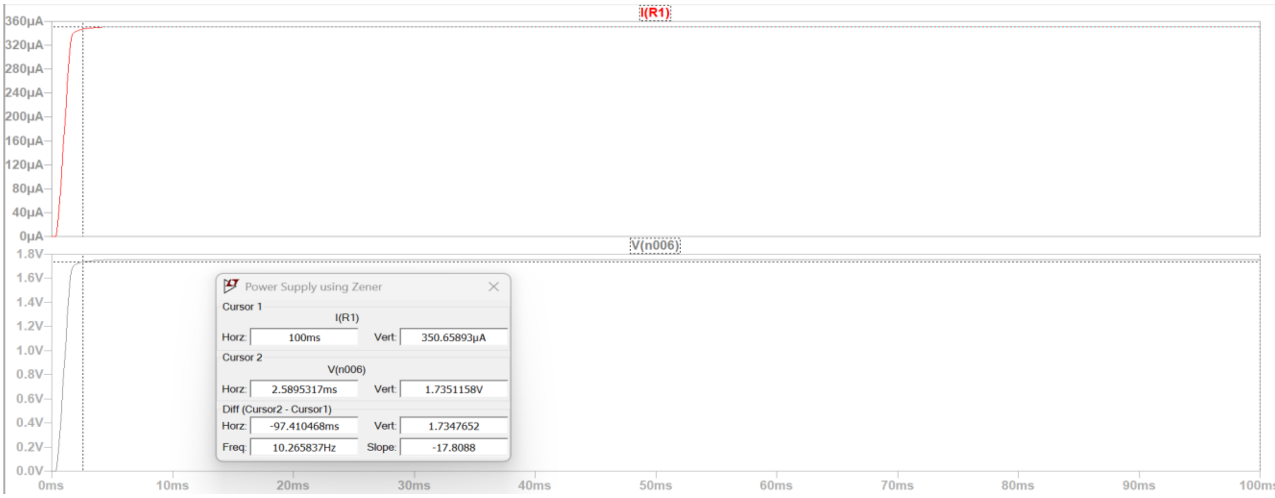

Figure 20: Output of complete circuit in LTSpice

Hard Ware Circuit

The Hard Ware prototype was assembled on a breadboard following the simulated schematic, using a stepdown transformer to interface with the AC mains. The rectification stage was implemented using 1N4007 Silicon diode in a bridge configuration, followed by a dual capacitor filter 47µF and 10µF to minimize the ripple voltage. The final regulation was achieved by biasing a 2.8V Zener diode through a 10kΩ resistor, with a subsequent voltage drop across a 470Ω resistor and a series connected diode to successfully deliver a measured 1.7V DC to the 5kΩ load.

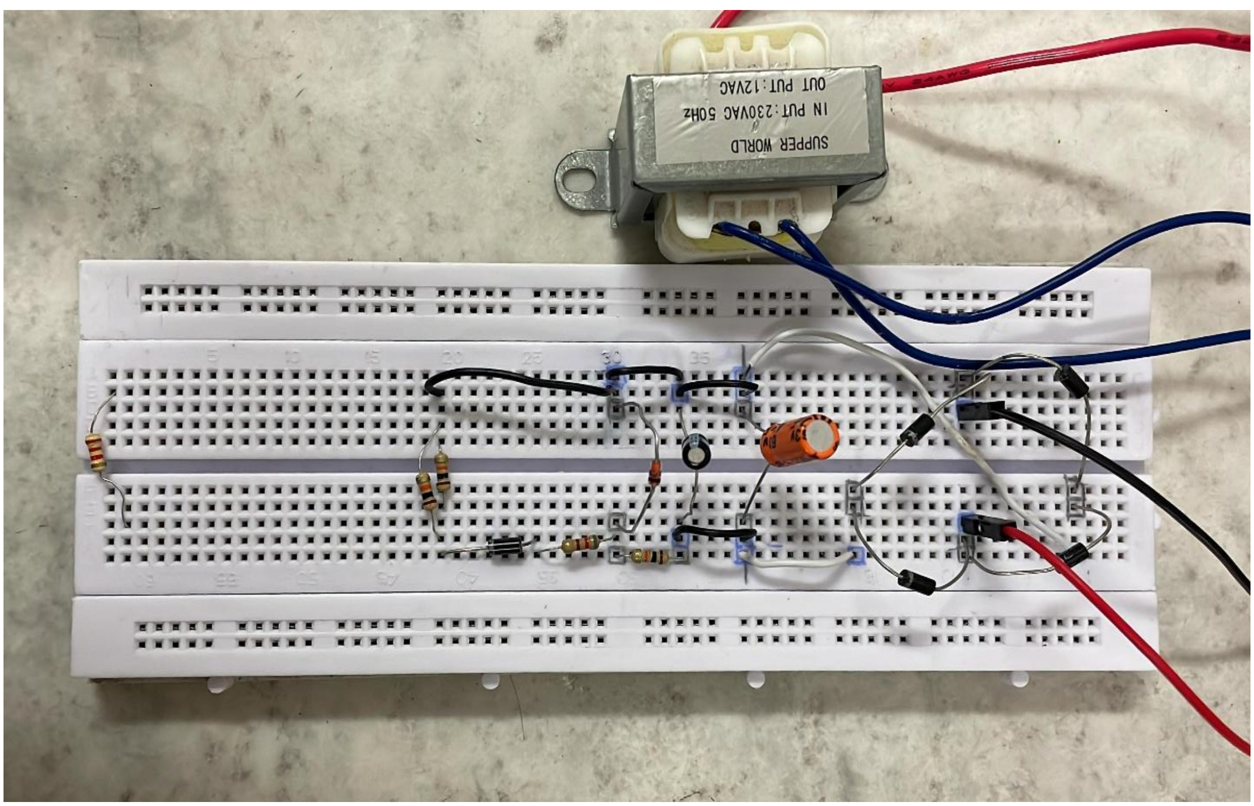

Figure 21: Complete Circuit on breadboard

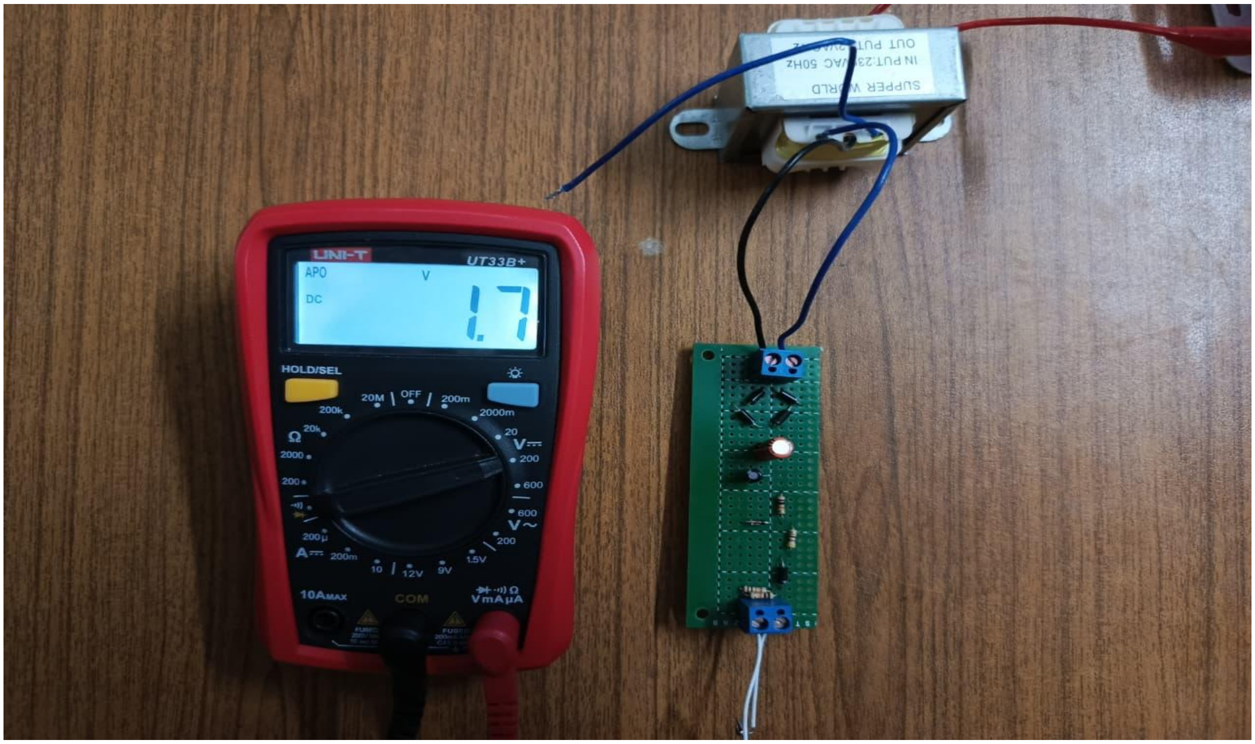

Figure 22: Circuit on veroboard

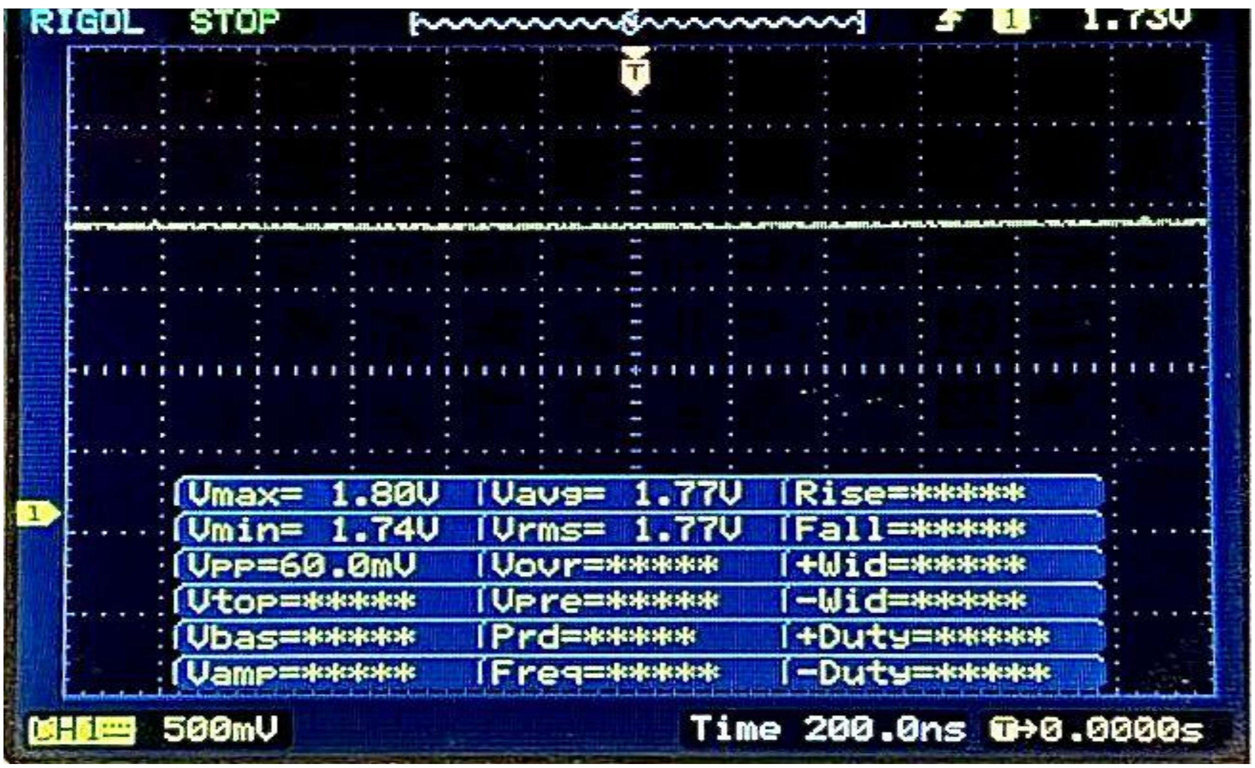

Figure 23: Output wave on Oscilloscope

Experimental Procedure

- Firstly, the circuit was made on a breadboard.
- A bridge rectifier was used at the output terminal of the step-down transformer.
- Then, the capacitor of rating 2200µF was connected to both output terminals of the bridge rectifier.
- After that, the positive terminal of the capacitor was connected to the input of the voltage regulator circuit, and the negative terminal was connected to the ground.
- At the end of the circuit, a resistor and an LED were connected in series with the regulated output terminal, and the other end of the LED was connected to the common ground shared with the capacitor.
- After completing the circuit on the breadboard, the secondary ends of the 12V step-down transformer were connected to the input terminals of the bridge rectifier.
- Now, the primary side of the transformer was connected to the supply board in the lab, and the output of the LED bulb was observed.
- The light emission from the LED bulb confirmed the working of the DC power supply.
- Finally, the voltage after each component on the breadboard was measured using the DMM, and the observed voltages were noted.

Discussion

The objective of this project was to design and implement a regulated DC power supply capable of delivering a stable 1.7V output across a 5kΩ load. Both simulation and hardware implementation were carried out to evaluate the performance of the designed circuit. The results obtained from LTspice simulation closely matched the theoretical calculations, confirming the correctness of the design approach. The transformer successfully stepped down the high AC mains voltage to a manageable level, and the bridge rectifier efficiently converted the AC signal into pulsating DC. However, as expected, a voltage drop of approximately 1.4V occurred across the rectifier diodes, which was accounted for during the design stage. The filter circuit, consisting of 47µF and 10µF capacitors in parallel, significantly reduced the ripple in the rectified output. The measured ripple percentage of approximately 3.39% indicates that the filtering stage was effective, although minor fluctuations still remained. These small ripples are typical in practical circuits due to non-ideal components and load variations. The Zener diode regulator played a crucial role in stabilizing the output voltage. By operating in the breakdown region, it maintained a nearly constant voltage despite variations in the input. The addition of a silicon diode in series allowed fine adjustment of the output voltage to the desired 1.7V level. The final measured current of approximately 0.35mA across the load confirms that the circuit performed as expected. In hardware implementation, slight deviations from simulation results were observed. These differences can be attributed to component tolerances, temperature effects, and non-ideal characteristics of real devices such as internal resistance of the transformer and leakage currents in capacitors. Despite these variations, the circuit successfully delivered a stable and regulated output, validating the overall design.

Conclusion

This project successfully demonstrated the design and implementation of a regulated DC power supply capable of providing a stable 1.7V output across a 5kΩ load. The combination of transformer, bridge rectifier, filter circuit, and Zener diode regulator effectively converted AC input into a smooth and regulated DC output. The agreement between theoretical calculations, simulation results, and experimental observations confirms the reliability of the design methodology. Although minor discrepancies were observed in practical implementation, the overall performance remained within acceptable limits. This experiment highlights the importance of each stage in a power supply system, particularly the role of filtering and regulation in ensuring voltage stability. The project also provided practical insight into real world limitations such as ripple, component tolerances, and voltage drops. In conclusion, the regulated power supply designed in this project meets the required specifications and demonstrates a simple yet effective approach for obtaining low voltage DC output for electronic applications.